Should Chart Buzz Decide the Room?

What 5,806 concerts say about sizing a tour · Matt Steines · November 2026 · github.com/msteines0312/concert-ticket-tracking

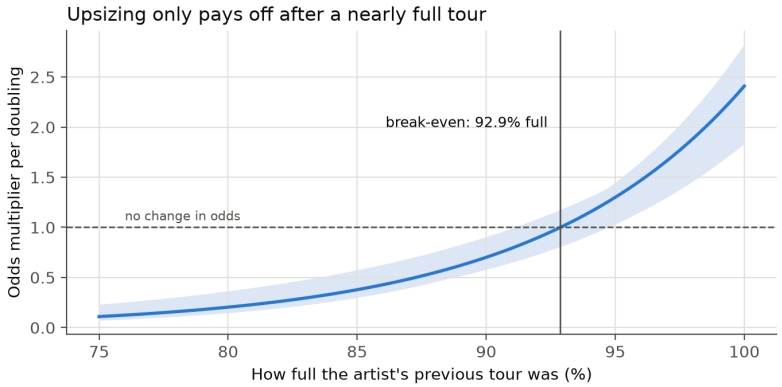
> **Short answer:** no. A fresh Billboard run barely improves a sellout forecast once you know the room size and the market. The more useful number is how full the artist's last tour was. Moving an act into a bigger room paid off after a tour that was about 93% full or better, and backfired below that.

The question

A talent buyer usually books an artist months before the show, sometimes while one of their songs is still climbing the charts. Too small a room leaves money on the table. Too big a room means empty seats, which hurts the artist and the venue. I wanted to know whether chart and streaming attention, measured six months out when a show would typically be booked, predicts which shows sell out. And if it doesn't, what should a buyer look at instead?

Finding 1: upsizing works only after a nearly full tour

For artists with a previous tour, I compared each show's room with that tour's typical room. When the last tour was 100% full, doubling the room multiplied the odds of a sellout by about 2.4. After a 95%-full tour the gain shrinks to 1.3, and after a 90%-full tour doubling *cuts* the odds to 0.7. The break-even is **92.9% full**.

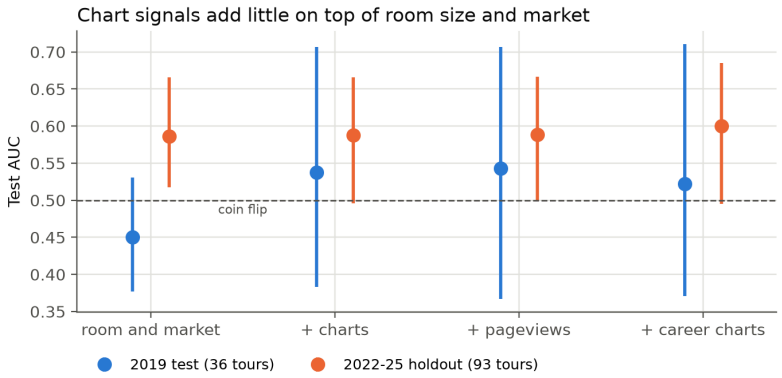


The shaded band is the range across 115 refits that each leave out one tour.

Halsey's Love and Power Tour is the clearest case. Her previous tour was 83% full, the new rooms were 2.4 times bigger, and 2 of 21 shows sold out. The pattern held when any single tour was left out (break-even between 91.1% and 94.7%) and when "sold out" was loosened to 97% full. Shrinking the room showed no effect either way.

Finding 2: chart buzz doesn't forecast sellouts

The raw numbers make charts look useful: artists with more chart activity do sell out more often. Most of that comes from bigger acts charting more and from later years selling out more. Once room size and market are in the model, the charts add almost nothing. I trained on 2017-18 tours and tested on tours the models never saw. On 2022-25 tours, adding chart signals left AUC at 0.587, where 0.5 is a coin flip. On 2019 tours it rose from 0.450 to 0.538, but with only 36 test tours the interval on that gain runs from -0.10 to +0.27. Wikipedia pageviews and Spotify streams did no better.



Each step adds a block of signals. Bars are 95% intervals from resampling whole tours.

How a buyer could use this

Before making an offer in a bigger room, ask one question: was the last tour at least 93% full? Looking back, 13 tours had at least half their shows upsized against that rule, and they sold out less often.

	Shows	Sold out
Upsized against the rule (13 tours)	211	70.1%
Every other show with a previous tour (106 tours)	2,716	79.3%

The check only needs two numbers a buyer already has: the last tour's fill rate and the size of the new room. A song climbing the charts can still be a good reason to look at an act. It just isn't evidence that the act can fill a bigger room.

How I tested it

- Attendance and capacity for each show come from Wikipedia box score tables for 240 tours (US and Canada, 2016-2026). I checked them against Pollstar's year-end charts, and 95% of 476 matched engagements agree within 1%.
- Chart, pageview and Spotify signals are all measured six months before the show, around when it would be booked. One test floods every later date with fake hype and confirms none of the features change.
- Shows on the same tour are near-copies, so each tour goes entirely to training or entirely to testing. Models train on 2017-18 tours and are scored on 2019 and on 2022-25.
- Same-show ticket price looked predictive at first. Dynamic pricing was reacting to demand, so the price was partly the outcome itself. The models use the previous tour's price instead.

Limits

- The sample leans toward big tours. A tour only gets in if it has a Wikipedia article with box office numbers, and 68% of shows in the data sold out. Club and theater acts may behave differently.
- The data is observational. The upsizing rule describes how past bookings turned out, and buyers may have known things the data doesn't. I'd use it as a check before an offer rather than as a forecast.
- When a show doesn't sell out, how full it gets is hard to predict. No model beat just guessing the average.

Sources. Wikipedia concert tour articles, 2015-2025 (CC BY-SA 4.0), with a per-article list in docs/wikipedia_sources.csv. Billboard Hot 100 and Billboard 200 charts via utdata/rwd-billboard-data (chart data © Billboard). Wikimedia Pageviews API (CC0). Spotify charts via Kaggle, 2017-2021. Pollstar year-end Top Concert Grosses 2019, 2022 and 2023, used for validation only (© Pollstar). BLS CPI-U for 2024 dollars.